

Privity

Both legs of a tokenized fund trade settle in **one Canton transaction**, and each counterparty sees only the parcel it is buying.

Settlement infrastructure for tokenized funds. Built for fund administrators, who hold the share register and produce NAV.



THE PROBLEM

Tokenized fund trades still settle **like paper**.

Cash moves on one rail. Units are typed into a register afterwards. A human reconciles the gap between them.

238 → 462

NAV ERRORS NOTIFIED TO THE CSSF, 2019 TO 2022

Investment compliance breaches **fell** over the same window. The failure mode that is growing is operational.

1 day

SEC PROPOSAL, SEPT 2026

Register posting moves to a one business day standard tied to the settlement cycle.

Sept 13

DTCC TOKENIZATION ON CANTON

The register is moving on chain whether or not settlement does.

CSSF figures reported by Deloitte Luxembourg from CSSF activity reports 2017 to 2022. SEC figures per Morgan Lewis analysis of the proposing release.

The regulator has already written our spec.

“for uncertificated securities, prompt posting and turnaround are effectively the same event”

SEC, on proposed Rules 17ad-2 and 17ad-10, September 2026

A transfer and a register update should be one event once securities stop being pieces of paper. Privity makes them literally one transaction.

Also in the proposal

- “item” extended to cover **blockchains and other distributed ledger-based platforms**
- An authorized-persons list you may only act on, which is the shape of our mandate model

Who it lands on

- **~200 registered transfer agents** in the exemption being rescinded, covering LP interests, DRIPs and open-end fund securities
- Small, named, regulated buyer set. High deal value.

Two properties that no other ledger offers **together**.

A public chain

Gives you atomic settlement. Also publishes every investor's position to every other investor. For a regulated fund that ends the conversation.

Off-ledger

Gives you confidentiality. Also keeps cash and register as two separate events, which is the problem itself.

Canton's sub-transaction privacy means one transaction commits across parties while each participant validates only the views its own parties are stakeholders on.

We did not take this on trust

Our first design had the buyer read the seller's holding to verify the trade. **The ledger rejected it**, because the buyer is not a stakeholder. Canton enforced the confidentiality property against our own code.

Calibrated disclosure, **not maximum secrecy.**

The obvious repair was to add the buyer as an observer on the seller's holding. That would have disclosed the seller's entire position. Both versions compile. Both settle correctly. Only a test written to falsify the claim tells them apart.

Hide everything

The buyer cannot verify what it is paying for.
Unusable.

Disclose the holding

The buyer learns the seller's whole book.
Pointless.

Disclose one parcel

The seller splits exactly the quantity being sold.
The buyer verifies that and learns nothing else.
The transfer then clears the disclosure.

The engineering is finding the floor between those two failures. For a secondary trade that floor is exactly one parcel.

PROOF, ON A REAL LEDGER

Four claims, each verified rather than asserted.

CLAIM	EVIDENCE	WHERE
Both legs commit together	1 transaction, 8 events	live participant
Buyer cannot read the seller's retained units	active contract query as buyer returns nothing	live participant
A short payment settles nothing	transaction rejected, parcel undelivered	test suite
NAV is checkable, not just signed	auditor recomputes digest and NAV, both true	live participant

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DAML TESTS PASSING

~2 min

LOCALNET FROM NOTHING TO HEALTHY

0

DAML-SCRIPT IN PRODUCTION DARS

NEED TO KNOW, READ BACK FROM THE LEDGER

Four parties. Four different views.

PARTY	WHY THEY SEE WHAT THEY SEE	SEES SELLER'S RETAINED UNITS
Seller	They are her own units	yes, as designed
Buyer	The claim under test	no, as required
Fund administrator	Stakeholder on the register. Cannot strike NAV otherwise	yes, as designed
Auditor	Entitled to the mandate and the attestation, not to positions	no, as required

Expected visibility is asserted **per role**. The administrator seeing the register is correct, not a leak. The test `testBuyerCannotSeeSellersRemainingPosition` fails the build if the buyer ever gains that visibility.

Three packages. The disclosure layer stands alone **on purpose.**

PACKAGE	CONTENTS	WHY SEPARATE
privity-disclosure	Mandates, windows, audit records	Zero fund logic, so any Canton app can depend on it
privity	Fund, register, cash, NAV, settlement	The application
privity-tests	Daml Script suite	Keeps test tooling out of production DARs

Both sides sign a mandate

If only the auditor signed, anyone could assert supervision over anyone. If only the subject signed, a firm could manufacture evidence of an audit that never happened.

Upgrade-safe, the hard way

Moving mandate validity to ledger time meant deleting a choice field. Canton refused the upload. The field is retained and ignored, so 1.1.0 vets as a valid upgrade of 1.0.0.

WHAT IS NOT DONE

Stated here rather than discovered by a judge.

the real gap

Zero practitioner interviews

Problem evidence is documentary and regulator-sourced, which moves problem validation and not market validation. The instrument and a named target list are built. A falsification test is committed in writing: if three ICP interviews say the timing gap is already handled, the wedge is wrong and the product moves.

scope, deliberately

Single synchronizer, LocalNet only

A Daml transaction can only consume contracts on one synchronizer, so atomicity is demonstrated in that topology, which is what the Global Synchronizer provides. The cross-synchronizer case is documented, needs no model change, and carries no settlement risk because no value moves until the atomic step.

Nothing deployed beyond LocalNet. No production integration with an admin system, which is the largest practical adoption risk.

A small, named, regulated buyer set.

The person

- **Pain owner:** fund accounting or transfer agency lead at a mid-tier administrator, 20 to 60 funds
- **Economic buyer:** head of operations
- **Influencer:** the auditor. Not the buyer.

The model

- **Per fund per year**, plus a platform fee
- Not per seat: the value is process containment, not user productivity
- Land one administrator, expand across its funds. One sale, thirty deployments

On the neighbour in this track

Alluvren solves the front-end redemption decisioning layer. Privity solves the back-end bilateral settlement and disclosure layer. They may coexist in the same fund workflow.

THE ASK

Three introductions, and one design review.

Introductions

Three fund administrators for validation interviews. This is the single highest-value thing anyone here can give us, because it is the one gap we cannot close alone.

A design review

Should `privity-disclosure` be upstreamed as an ecosystem primitive? Every Canton team building a multi-party workflow hits this wall, and most will reach for the observer field.

Operator guidance

What a participant operator considers acceptable for holding a share register in production.

After the competition the disclosure package is intended for Apache-2.0 release, so it can be adopted rather than admired.

Settlement that cannot half-complete. Disclosure that cannot overreach.

DEMO	https://zkasuran.github.io/privity-demo/
CODE	https://github.com/zkasuran/privity
TRACK	Investment Infrastructure: Funds, DAOs & Governance Tools

Every figure in this deck came out of a real Canton participant or a cited public source.
Where something is incomplete, the deck says so.

AI assistance (Claude, Anthropic) was used in developing this project. Design decisions, review and verification are the author's. Verified before submitting: dpm build --all succeeds for three packages, dpm test passes 18 of 18, both production DARs vet on a live Canton participant, and the demo URL returns 200 anonymously.